Quantum State Discrimination with Bosonic Channels and Gaussian States

Course: PH3203 Advanced Quantum Mechanics
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A Term Paper Review of Chapters 1–3[1]

Based on the MIT Doctoral Thesis by Si Hui Tan (2010)

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Abstract

This report comprehensively reviews the first three chapters of Si Hui Tan's MIT doctoral thesis, "Quantum State Discrimination with Bosonic Channels and Gaussian States."

We begin by establishing the foundational concepts of quantum state discrimination, including the notion of quantum states, measurement postulates, and the Positive Operator Valued Measure (POVM) formalism. We then explore the two primary discrimination strategies – minimum-error discrimination (Helstrom bound) and unambiguous discrimination (IDP limit) –along with practical approaches such as the pretty-good measurement and the no-measurement strategy. We highlight the central role of state discrimination in quantum information theory through its applications in quantum key distribution, entanglement characterization, quantum error correction, and quantum algorithms.

We further develop the continuous variable quantum information framework, including the quantization of the electromagnetic field, bosonic mode operators, the symplectic structure of phase space, Gaussian transformations, and phase-space representations such as the Wigner function and characteristic function. We provide detailed derivations of coherent states, squeezing operators, two-mode squeezing, and Gaussian EPR states. We introduce Williamson's theorem as a key tool for decomposing covariance matrices and include a complete step-by-step proof.

Finally, we examine the distinguishability of Gaussian states using fidelity and the quantum Chernoff bound. We present a numerical implementation using QuTiP to validate the Helstrom bound and the quantum Chernoff bound for multi-copy state discrimination, demonstrating that the quantum Chernoff bound provides an asymptotically tight upper bound on the minimum error probability.

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1 Introduction

1.1 The notion of a quantum state

A quantum state represented by a positive Hermitian operator $\hat{\rho}$ with unit trace, contains all physically relevant information about a quantum system. If two quantum systems are prepared in identical states, any measurement performed on them will yield identical outcome probabilities, making the different preparation methods observationally equivalent. Conversely, if the systems are in different quantum states, there necessarily exists at least one measurement that produces different probability distributions, allowing their preparations to be distinguished. The systematic process of designing and performing measurements to identify which of several possible states (and thus which preparation) a given system is in is called quantum state discrimination.

1.2 Measuring a quantum state

While the quantum state itself is an abstract mathematical object, measurements provide it with concrete operational meaning. The outcomes of quantum measurements are always classical, such as a photodetector click count or an ion's position in a trap, and these outcomes behave as random variables. By performing the same measurement many times on identically prepared quantum systems, one can reconstruct the statistical distribution of outcomes. If this process is repeated for every possible measurement, the complete set of outcome probability distributions fully characterizes the quantum state. Thus, the state is not directly observable, but its empirical reality is revealed through the statistics of measurement outcomes, bridging the gap between abstract formalism and experimental practice. The formal relationship between quantum states and measurements is captured by the postulates of standard quantum measurement theory. These postulates can be summarized as follows:

1. Every physical observable corresponds to a Hermitian operator $\hat{X}$ with a spectral decomposition
$$\hat{X} = \sum_j \lambda_j |j\rangle\langle j| \tag{1.1}$$
where λ_j are the possible measurement outcomes.

2. The projectors
$$P_j = |j\rangle\langle j| \tag{1.2}$$
associated with the eigenstates form a complete set, satisfying
$$\sum_j P_j = \mathbb{I} \tag{1.3}$$

and are orthogonal such that

$$P_i P_j = \delta_{ij} P_i \tag{1.4}$$

3. A measurement of $\hat{X}$ yields one of its eigenvalues λ_j as the result.

4. If the outcome λ_j is obtained, the system collapses to the post-measurement state

$$\hat{\rho}_j = \frac{P_j \hat{\rho} P_j}{\mathrm{tr}(P_j \hat{\rho} P_j)} \tag{1.5}$$

5. The probability of obtaining the outcome λ_j is given by

$$p_j = \mathrm{tr}(\hat{\rho} P_j) \tag{1.6}$$

6. When the measurement is performed but the outcome is not recorded, the post-measurement state is described by the density operator

$$\hat{\rho}' = \sum_j P_j \hat{\rho} P_j \tag{1.7}$$

representing an average over all possible outcomes.

It is important to note that in stating the postulates, the Hilbert space has been assumed to be finite and discrete solely for the sake of notational simplicity. In principle, these postulates apply equally to states with infinite-dimensional Hilbert spaces, which is precisely the case for continuous variable systems. The postulates make it clear that the measurement process is fundamentally random, with the outcome able to take a range of values. The number of possible measurement outcomes is equal to the dimension of the Hilbert space of the state. The projection operators P_j used in this formulation are referred to as projective measurements or von Neumann measurements. However, it turns out that the requirement that measurement operators be strictly projective can be relaxed while still maintaining a meaningful statistical distribution for the outcomes. In particular, Postulate 3 provides an algorithm for generating outcome probabilities.

If we replace the projective measurement operator P_j with a positive operator Π_j that may not necessarily be a projection, the probability of obtaining outcome j is given by

$$p_j = \mathrm{tr}(\hat{\rho} \Pi_j) \tag{1.8}$$

Since Π_j is a positive operator, $p_j \geq 0$, and if we further require that the set of such operators for a finite set of outcomes is complete, meaning $\sum_j \Pi_j = \mathbb{I}$, then the quantities p_j can be interpreted as probabilities for the outcomes of the measurement described by

the operator set $\{\Pi_j\}$. Such a set of measurement operators is called a positive operator valued measure, or POVM. A key feature of POVMs is that their elements do not need to be orthogonal. Consequently, the number of possible outcomes can be larger than the dimension of the Hilbert space of the state being measured, offering greater flexibility in measurement design.

Using POVMs, we can replace the postulates of the quantum theory of standard measurements with the following more flexible ones:

1. Measurement operators are positive operators $\Pi_j \geq 0$ where the set of all such operators is complete: $\sum_j \Pi_j = \mathbb{I}$. The set of such operators is called a POVM. The elements of the POVM can be written in terms of generalized measurement operators A_j, where
$$\Pi_j = A_j^\dagger A_j \tag{1.9}$$
The generalized measurement operators can be non-Hermitian, though the elements of the POVM must be Hermitian by construction.

2. Each element of the POVM set represents a possible outcome of the measurement.

3. The state of the system after the measurement is
$$\hat{\rho}_j = \frac{A_j \hat{\rho} A_j^\dagger}{\mathrm{tr}(A_j \hat{\rho} A_j^\dagger)} \tag{1.10}$$

4. The probability that the outcome j (indexing the measurement operator Π_j) is found is
$$p_j = \mathrm{tr}(\Pi_j \hat{\rho}) \tag{1.11}$$

5. If we perform the measurement but do not record the results, the post-measurement state is described by the density operator
$$\hat{\rho}' = \sum_j A_j \hat{\rho} A_j^\dagger \tag{1.12}$$

The POVM formalism described above provides a way to describe classical information obtained from a quantum state by way of a measurement. State discrimination is the process of processing this information to determine, to the best of our knowledge, which quantum state we possess. The fundamental question of how well we can discriminate between quantum states lies at the heart of quantum mechanics. When the set of possible states are orthogonal to one another, perfect discrimination is always possible. This can be achieved by choosing a set of projection operators that correspond exactly to the

set of possible states. However, when the possible states are non-orthogonal, perfect discrimination is impossible. In such cases, we must define optimal discrimination with respect to some figure of merit, though finding this optimal discrimination is often non-trivial. The chapter will next set up the problem of state discrimination and describe two commonly used strategies: minimum-error discrimination and unambiguous discrimination. Following that, some interesting well-known measurements that do not fall into these two categories, such as the pretty-good measurement and the no-measurement strategy, will be discussed.

1.3 State Discrimination

Consider an ensemble of quantum systems such that each system is prepared in one of N possible states, say $\hat{\rho}_1, \ldots, \hat{\rho}_N$, with a priori probabilities $p_1, \ldots, p_N$ respectively, where $\sum_{j=1}^{N} p_j = 1$. In general, the states can be pure or mixed, and they can be either discrete variable states or continuous variable states. The observer performing state discrimination has full knowledge of the ensemble being prepared, meaning he knows the possible states and their prior probabilities. However, he does not know which one of the N states he possesses, and he must make a measurement to determine this. State discrimination describes how he should design and perform these measurements to optimally identify the unknown state.

There are two commonly used strategies for state discrimination. The first is discrimination with minimum error, attributed to Helstrom, who studied distinguishing between two given quantum states. Helstrom found an optimum measurement strategy for both mixed and pure states that minimizes the probability of error. This optimum measurement strategy is known as the Helstrom measurement, and the resulting minimum error is called the Helstrom bound. The general problem of discriminating between $N > 2$ states with minimum error is considerably more difficult. Although the necessary and sufficient conditions for an optimal measurement have been found, how to implement such an optimal measurement remains an open problem, except for a few special cases.

The second strategy is unambiguous discrimination, first proposed by Ivanovic. This strategy allows perfect discrimination between two non-orthogonal states, but with the caveat that a non-conclusive (inconclusive) answer is permitted. The specific case of discriminating between two pure non-orthogonal states was solved by Dieks and Peres. However, the general case of unambiguously discriminating between N pure states remains an open problem. It has been shown by Chefles that unambiguous discrimination is only possible if the states to be distinguished are linearly independent. Additionally, there are active research areas focused on extending unambiguous discrimination strategies to

mixed states.

Having described the two strategies briefly, let us now discuss each of them in more detail.

1.3.1 State Discrimination with Minimum Error

In the minimum error discrimination strategy, we always reach a conclusion about which state was prepared, though errors in this conclusion are inevitable. The goal of the optimal strategy is to minimize the detection error. To formulate this as a definite optimization problem, we must define what constitutes a detection error.

Let Π_j be the POVM element corresponding to the state $\hat{\rho}_j$ being prepared. According to the postulates of quantum measurement theory, if the actual prepared state is $\hat{\rho}$, the probability of guessing that the state is $\hat{\rho}_j$ is $\mathrm{tr}(\hat{\rho}\Pi_j)$. Since the state $\hat{\rho}_j$ occurs with a priori probability p_j, the overall probability of a correct guess is given by

$$P_{\mathrm{corr}} = \sum_{j=1}^{N} p_j \,\mathrm{tr}(\hat{\rho}_j \Pi_j) \tag{1.13}$$

The detection error is then defined as

$$P_{\mathrm{err}} = 1 - P_{\mathrm{corr}} \tag{1.14}$$

To minimize the error, we must minimize P_{err} over all possible sets of POVM elements $\{\Pi_j\}$ for $j = 1, \ldots, N$ subject to the constraints $\Pi_j \geq 0$ and $\sum_{j=1}^{N} \Pi_j = \mathbb{I}$. The POVM that achieves this minimization is called the optimal measurement, and the resulting error is the minimum error. The special case of distinguishing single copies of two mixed states will be derived later in Chapter 3.

1.3.2 Unambiguous State Discrimination

Unlike discrimination with minimum error, in unambiguous discrimination we are not permitted to make measurement errors. However, this strategy does not succeed 100% of the time, and this trade-off is necessary to achieve error-free discrimination.

To illustrate how unambiguous state discrimination works, we consider a simple example involving two pure states $|\psi_0\rangle$ and $|\psi_1\rangle$ occurring with equal a priori probabilities, given by

$$|\psi_0\rangle = \cos\theta|0\rangle + \sin\theta|1\rangle \tag{1.15}$$

$$|\psi_1\rangle = \cos\theta|0\rangle - \sin\theta|1\rangle \tag{1.16}$$

These states are non-orthogonal with overlap $\langle\psi_0|\psi_1\rangle = \cos^2\theta - \sin^2\theta = \cos(2\theta)$.

A POVM for distinguishing between these states can be constructed with elements

$$\Pi_0 = a_0(\sin\theta|0\rangle + \cos\theta|1\rangle)(\sin\theta|0\rangle + \cos\theta|1\rangle) \tag{1.17}$$

$$\Pi_1 = a_1(\sin\theta|0\rangle - \cos\theta|1\rangle)(\sin\theta|0\rangle - \cos\theta|1\rangle) \tag{1.18}$$

$$\Pi_2 = \mathbb{I} - \Pi_0 - \Pi_1 \tag{1.19}$$

where the coefficients are chosen such that

$$\langle\psi_0|\Pi_1|\psi_0\rangle = \langle\psi_1|\Pi_0|\psi_1\rangle = 0 \tag{1.20}$$

and $0 \leq a_0, a_1 \leq 1$. When outcome 0 is obtained, we can confidently conclude the state is $|\psi_0\rangle$; when outcome 1 is obtained, we can confidently conclude the state is $|\psi_1\rangle$. However, if outcome 2 occurs, no definite conclusion can be made.

To evaluate how effective this unambiguous discrimination strategy is, we need to determine the probability that the measurement yields an inconclusive result. A lower probability of inconclusive outcomes indicates a better discrimination scheme. Therefore, we now derive this probability and then optimize it over the available parameters. We begin by computing the conclusive probabilities. we first compute the conclusive probabilities. The probability of obtaining outcome 0 when $|\psi_0\rangle$ is sent is

$$\langle\psi_0|\Pi_0|\psi_0\rangle = a_0|\langle\psi_0|(\sin\theta|0\rangle + \cos\theta|1\rangle)|^2 \tag{1.21}$$

The overlap is

$$\langle\psi_0|(\sin\theta|0\rangle + \cos\theta|1\rangle) = \cos\theta\sin\theta + \sin\theta\cos\theta = \sin(2\theta) \tag{1.22}$$

Thus,

$$\langle\psi_0|\Pi_0|\psi_0\rangle = a_0\sin^2(2\theta) \tag{1.23}$$

Similarly, for $|\psi_1\rangle$,

$$\langle\psi_1|\Pi_1|\psi_1\rangle = a_1\sin^2(2\theta) \tag{1.24}$$

Since the prior probabilities are equal ($p_0 = p_1 = 1/2$), the overall probability of a conclusive result is

$$P_{\text{conc}} = \frac{1}{2}a_0 \sin^2(2\theta) + \frac{1}{2}a_1 \sin^2(2\theta) \tag{1.25}$$

Completeness of the POVM and positivity of Π_2 imposes constraints on a_0 and a_1. The optimal choice that maximizes the conclusive probability (or equivalently minimizes the inconclusive probability) is found to be $a_0 = a_1 = 1/(1 + \cos(2\theta))$. Substituting this value, we obtain

$$\langle\psi_0|\Pi_0|\psi_0\rangle = \frac{\sin^2(2\theta)}{1 + \cos(2\theta)} = \frac{1 - \cos^2(2\theta)}{1 + \cos(2\theta)} = 1 - \cos(2\theta) \tag{1.26}$$

Thus, the probability of obtaining an inconclusive result when $|\psi_0\rangle$ is sent is $1 - (1 - \cos(2\theta)) = \cos(2\theta)$. The same holds for $|\psi_1\rangle$. Therefore, the overall probability of an inconclusive result is

$$P(\text{inconclusive result}) = \cos(2\theta) \tag{1.27}$$

This minimum value is known as the Ivanovic-Dieks-Peres (IDP) limit. Analytical solutions for the IDP limit are not known for the general case of $N > 2$ pure states, though the solution for three pure states has been derived. Various numerical techniques have also been developed to calculate the IDP limit in the general case.

1.3.3 Pretty-Good Measurement

More often than not, in minimum-error state discrimination, it is not possible to find the optimal measurement. In such cases, the pretty-good measurement (PGM), also known as the square-root measurement, is—as its name suggests—a good measurement strategy to pursue. For a given set of N possible states, we can always construct the POVM elements of the PGM as

$$\hat{\Pi}_i = p_i \hat{\rho}^{-1/2} \hat{\rho}_i \hat{\rho}^{-1/2} \text{ for } i = 1, \ldots, N \tag{1.28}$$

where

$$\hat{\rho} = \sum_i p_i \hat{\rho}_i \tag{1.29}$$

and p_i is the probability of the ith state $\hat{\rho}_i$ occurring. The POVM elements defined in this way satisfy both positivity and completeness. Although the PGM is not the optimal measurement in the general case, it turns out to be optimal for many of the cases in which the minimum-error measurement is known. These results have been summarized in the literature.

1.3.4 Other Measurement Strategies

An interesting state discrimination strategy is the no-measurement strategy. For some sets of states, the optimal measurement is to make no measurements at all and simply guess the state that is most likely. A practical example where the no-measurement strategy applies is distinguishing a state from random noise: if the signal-to-noise ratio is sufficiently low, one should always guess that the received state was random noise.

Additionally, there exist mixed strategies that allow inconclusive results while minimizing the probability of detection error, as considered by Chefles and Barnett. The suggested potential use of such mixed strategies is to provide added flexibility for an eavesdropper to "listen in" on a cryptographic protocol.

Having discussed some of the strategies for state discrimination, important questions still remain: How does state discrimination help our understanding of quantum mechanics, and how can it be used in quantum protocols? The next section will address these questions.

1.4 The Role of State Discrimination in Quantum Information Theory

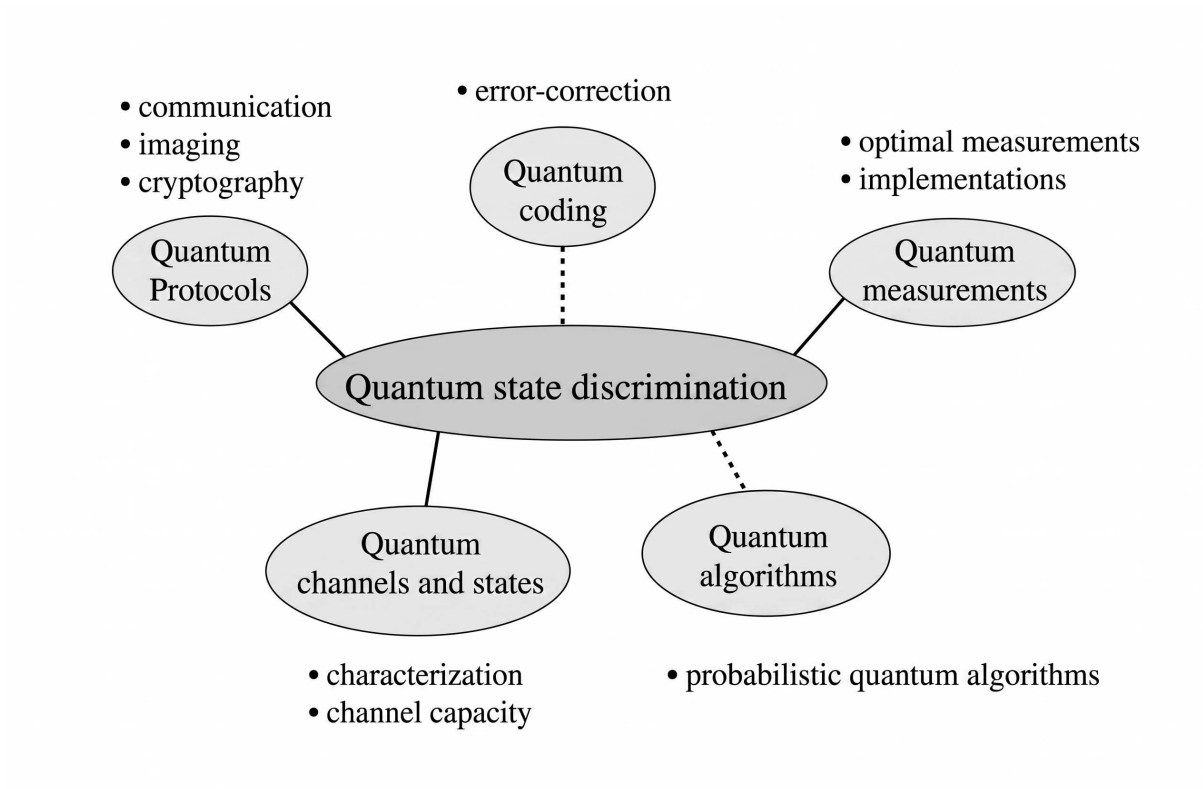

Figure 1: An overview of the central role of quantum state discrimination in quantum information theory.[1]

Quantum state discrimination lies at the heart of quantum computation and communication. The specific strategy of state discrimination applied depends on the task in quantum information theory to be solved. For example, using state discrimination techniques, we can show that quantum mechanics does not permit perfect discrimination between nonorthogonal states $|\psi_0\rangle$ and $|\psi_1\rangle$—a result that is also directly implied by the no-cloning theorem.

To demonstrate this, assume we have a POVM with two elements Π_0 and Π_1. For perfect discrimination, we require that

$$\Pi_0|\psi_1\rangle = 0 \tag{1.30}$$

$$\Pi_1|\psi_0\rangle = 0 \tag{1.31}$$

This means that if outcome 0 is obtained, we deduce the state was $|\psi_0\rangle$, and if outcome 1 is obtained, we deduce the state was $|\psi_1\rangle$. The POVM operators must also satisfy the completeness relation

$$\Pi_0 + \Pi_1 = \mathbb{I} \tag{1.32}$$

To derive the necessary condition for perfect discrimination, we multiply the above equation on the left by $\langle\psi_0|$ and on the right by $|\psi_1\rangle$:

$$\langle\psi_0|(\Pi_0 + \Pi_1)|\psi_1\rangle = \langle\psi_0|\mathbb{I}|\psi_1\rangle \tag{1.33}$$

Distributing the left-hand side gives

$$\langle\psi_0|\Pi_0|\psi_1\rangle + \langle\psi_0|\Pi_1|\psi_1\rangle = \langle\psi_0|\psi_1\rangle \tag{1.34}$$

Now, from the perfect discrimination conditions, we have $\Pi_0|\psi_1\rangle = 0$. Therefore, the first term $\langle\psi_0|\Pi_0|\psi_1\rangle = 0$. For the second term, note that Π_1 is a positive operator. If $\Pi_1|\psi_0\rangle = 0$, then taking the adjoint gives $\langle\psi_0|\Pi_1 = 0$. However, this does not directly imply $\langle\psi_0|\Pi_1|\psi_1\rangle = 0$. In standard unambiguous discrimination, the POVM elements are constructed such that Π_1 projects onto a subspace orthogonal to $|\psi_0\rangle$, which indeed forces $\langle\psi_0|\Pi_1|\psi_1\rangle = 0$. Under this construction, we have,

$$0 + 0 = \langle\psi_0|\psi_1\rangle \tag{1.35}$$

Thus,

$$\langle\psi_0|\psi_1\rangle = 0 \tag{1.36}$$

This condition can only be satisfied if the states $|\psi_0\rangle$ and $|\psi_1\rangle$ are orthogonal. Conse-